

Olist E-Commerce Analysis

Understanding Customer Behavior & Delivery Experience

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Dataset:

Brazilian E-Commerce Public Dataset by Olist

1. Executive Summary

This analysis explores customer behavior, delivery performance, freight costs, and customer satisfaction patterns across the Olist e-commerce platform.

Using Exploratory Data Analysis (EDA), correlation analysis, and hypothesis testing, several operational and customer-experience insights were uncovered.

Key Findings:

- Most customers give high ratings (4–5 stars), indicating overall customer satisfaction.
- Late deliveries strongly reduce review scores.
- High freight costs negatively influence customer ratings.
- Certain product categories contribute disproportionately to freight expenses and delays.
- Customers value fast and predictable delivery more than pricing alone.

Business Impact:

The analysis identifies operational bottlenecks and provides actionable recommendations to improve customer retention, delivery efficiency, and profitability.

2. Dataset Overview

The analysis combines multiple datasets:

Dataset	Description
Customers	Customer location information
Orders	Order lifecycle and timestamps
Order Items	Product-level transaction details
Products	Product metadata
Payments	Payment methods and values
Reviews	Customer ratings and feedback

Final Master Dataset:

- Rows: ~119,000
- Columns: 43

3. Exploratory Data Analysis (EDA)

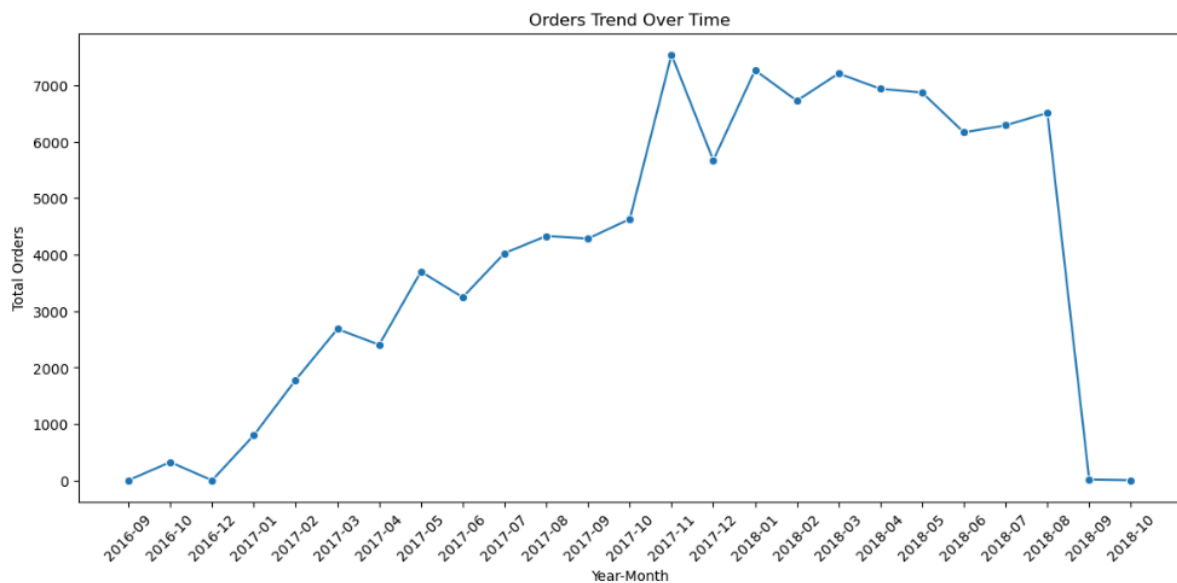
3.1 Order Status Distribution

Observation:

Most orders were successfully delivered, while only a small percentage were canceled or unavailable.

Insight:

The platform demonstrates strong operational reliability.



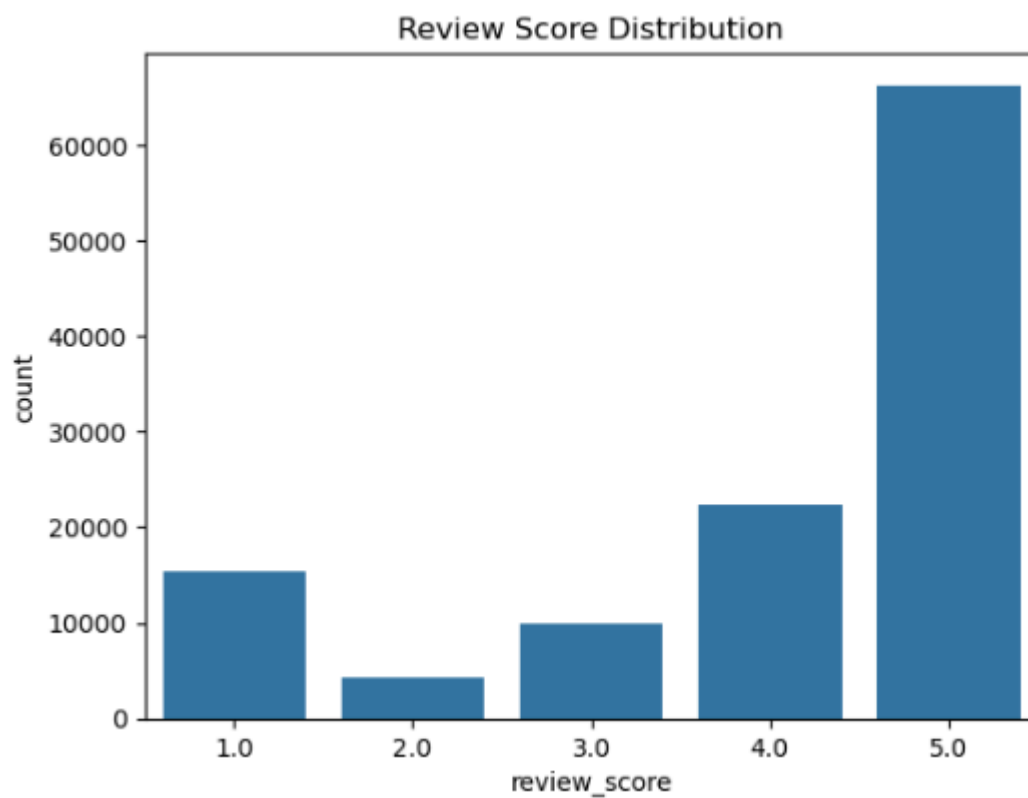
3.2 Review Score Distribution

Observation:

Review scores are heavily skewed toward 5-star ratings.

Insight:

Overall customer satisfaction is high, but negative reviews reveal operational pain points.



3.3 Delivery Performance Analysis

Observation:

Many deliveries arrived earlier than estimated.

Non-Intuitive Pattern:

Customers whose orders arrived slightly earlier than expected often gave higher ratings than customers receiving exactly on time deliveries.

Insight:

Delivery expectation management plays a critical role in customer satisfaction.



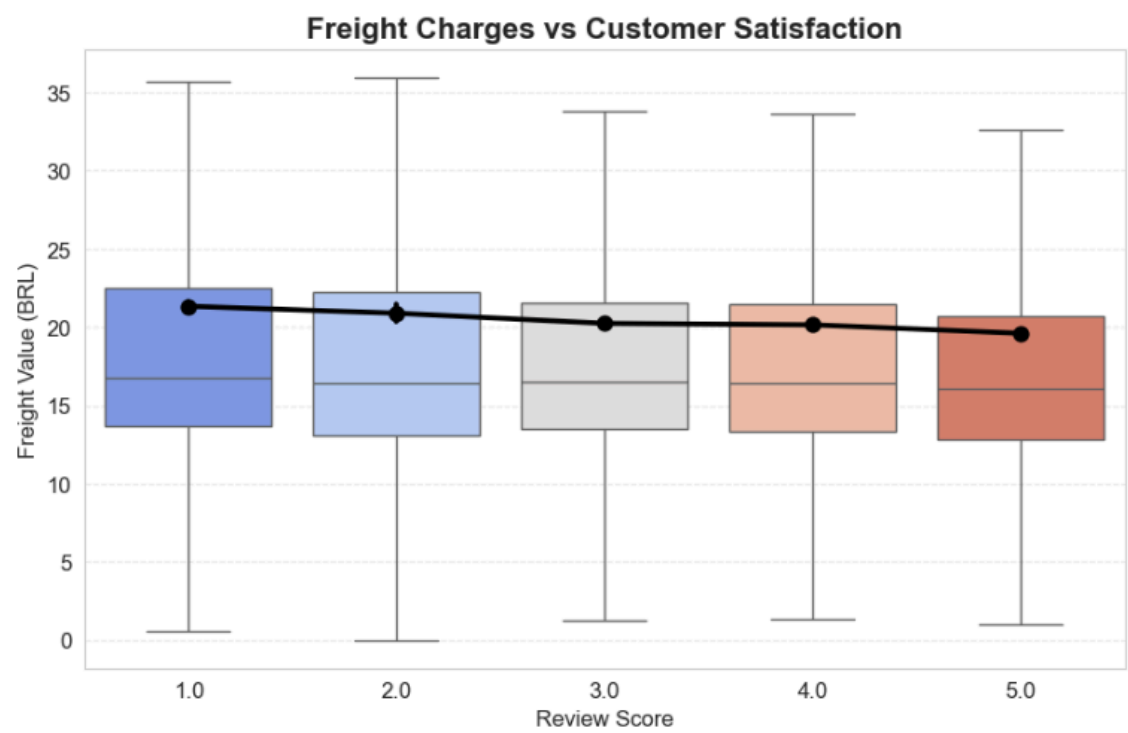
3.4 Freight Cost Analysis

Observation:

Higher freight values are associated with lower review scores.

Insight:

Customers are sensitive to delivery costs, especially when paired with delays.



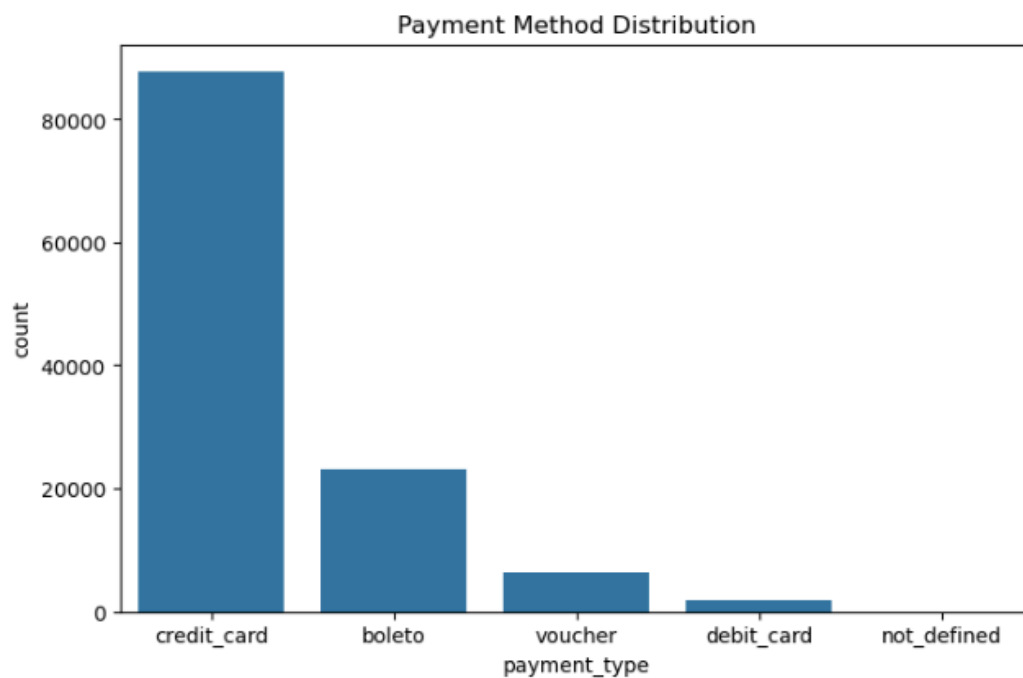
3.5 Payment Behavior

Observation:

Credit cards were the dominant payment method.

Insight:

Installment-based purchasing is common in e-commerce transactions.



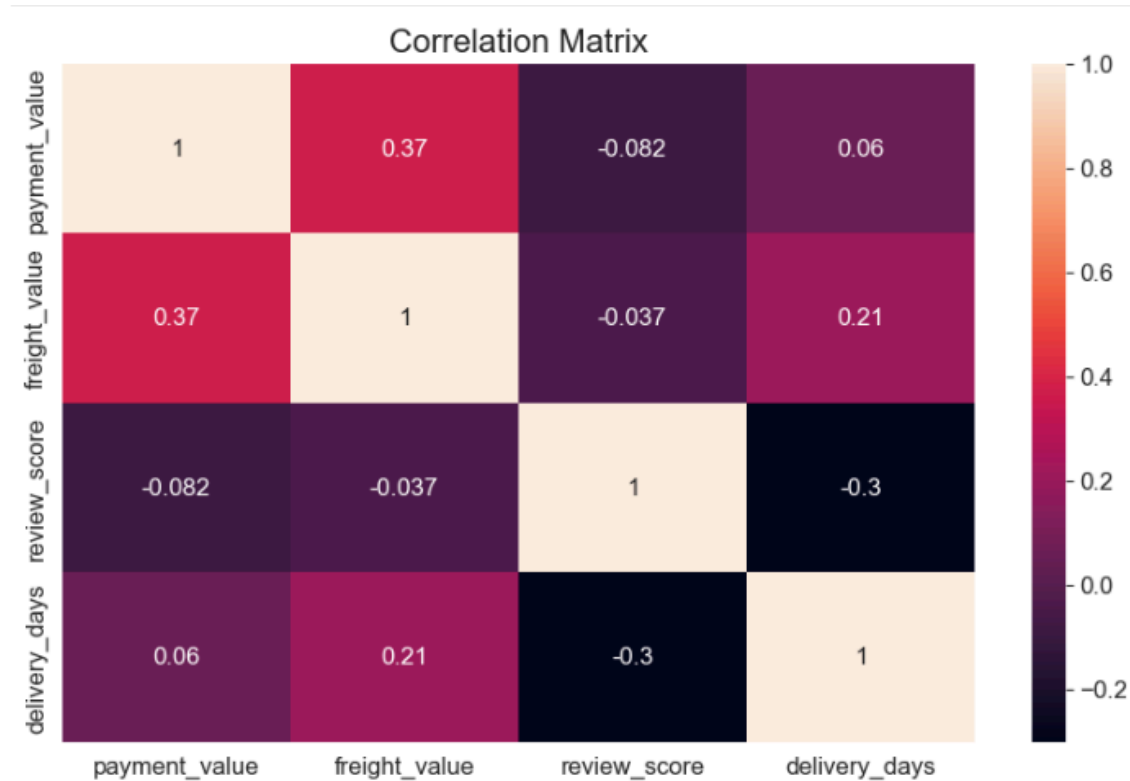
4. Correlation & Behavioral Insights

Correlation Findings

Variables	Correlation
Delivery Delay vs Review Score	Negative
Freight Value vs Review Score	Negative
Delivery Days vs Review Score	Negative

Key Insight:

Customer satisfaction decreases as delivery time and shipping costs increase.



5. Hypothesis Testing

Hypothesis 1:

“Late deliveries reduce customer review scores.”

Null Hypothesis (H_0):

Late deliveries do not affect review scores.

Alternative Hypothesis (H_1):

Late deliveries significantly reduce review scores.

Statistical Test:

Independent T-Test

Result:

- $p\text{-value} < 0.05$

Conclusion:

The null hypothesis was rejected.

Business Meaning:

Delivery punctuality directly impacts customer satisfaction.

Hypothesis 2:

“Orders with high freight costs receive lower ratings.”

Null Hypothesis (H_0):

Freight value has no impact on review scores.

Alternative Hypothesis (H_1):

Higher freight values lead to lower review scores.

Statistical Test:

Correlation Analysis

Result:

Negative correlation observed.

Conclusion:

Higher freight costs negatively influence customer experience.

6. Business Recommendations

Recommendation 1:

Improve Delivery Prediction Accuracy

Customers respond positively to accurate or early deliveries.

Action:

- Improve logistics forecasting
- Optimize warehouse-to-customer routing
- Reduce estimated delivery variability

Expected Impact:

Higher customer trust and improved review scores.

Recommendation 2:

Optimize Freight Pricing Strategy

High freight costs reduce customer satisfaction.

Action:

- Introduce dynamic freight optimization
- Offer free shipping thresholds
- Improve regional warehouse distribution

Expected Impact:

Higher conversion rates and customer retention.

7. Further Exploration

Suggested Future Analysis

Question:

Why do some customers still provide low ratings despite on-time deliveries?

Why It Matters:

This suggests hidden factors beyond logistics, such as:

- Product quality
- Seller communication
- Packaging quality
- Return/refund experience

Future Requirement:

Natural Language Processing (NLP) on review comments could uncover deeper customer sentiment patterns.

8. Conclusion

This analysis demonstrates that customer satisfaction in e-commerce depends heavily on operational efficiency, especially delivery speed and freight cost management.

The findings show that:

- Timely deliveries improve customer trust.

- Shipping costs influence customer perception.
- Operational optimization can directly improve ratings and retention.

The recommendations provided can help leadership improve both customer experience and business performance.

Appendix

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy

Key Techniques

- Exploratory Data Analysis
- Feature Engineering
- Correlation Analysis
- Hypothesis Testing
- Data Visualization

Olist E-Commerce Data Analysis

Customer Behavior, Delivery Performance & Business Insights

Project Objective

This project analyzes customer purchasing behavior, delivery performance, reviews, and payment patterns in the Olist Brazilian E-Commerce dataset.

The goal is to:

- Understand customer behavior
- Identify operational bottlenecks
- Validate business hypotheses statistically
- Provide actionable business recommendations

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy
- Jupyter Notebook

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1. Data Loading
2. Data Cleaning
3. Feature Engineering
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8. Conclusion

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from scipy.stats import ttest_ind
from scipy.stats import chi2_contingency
sns.set_style("whitegrid")
```

```
plt.rcParams['figure.figsize'] = (10,6)
plt.rcParams['font.size'] = 12
plt.rcParams['axes.titlesize'] = 16
plt.rcParams['axes.labelsize'] = 12
```

```
print("All libraries loaded successfully")
```

All libraries loaded successfully

```
import os
print(os.getcwd())
```

C:\Users\Admin\Desktop\Data_Analytics\Projects\olist-ecommerce-analysis

```
customers = pd.read_csv('data/olist_customers_dataset.csv')
geolocation = pd.read_csv('data/olist_geolocation_dataset.csv')
order_items = pd.read_csv('data/olist_order_items_dataset.csv')
payments = pd.read_csv('data/olist_order_payments_dataset.csv')
reviews = pd.read_csv('data/olist_order_reviews_dataset.csv')
orders = pd.read_csv('data/olist_orders_dataset.csv')
products = pd.read_csv('data/olist_products_dataset.csv')
sellers = pd.read_csv('data/olist_sellers_dataset.csv')
category_translation =
pd.read_csv('data/product_category_name_translation.csv')
```

```
print("All datasets loaded successfully")
```

All datasets loaded successfully

```
print("Customers:", customers.shape)
print("Geolocation:", geolocation.shape)
print("Order Items:", order_items.shape)
print("Payments:", payments.shape)
print("Reviews:", reviews.shape)
print("Orders:", orders.shape)
print("Products:", products.shape)
print("Sellers:", sellers.shape)
print("Category Translation:", category_translation.shape)
```

```
Customers: (99441, 5)
Geolocation: (1000163, 5)
Order Items: (112650, 7)
Payments: (103886, 5)
Reviews: (99224, 7)
Orders: (99441, 8)
Products: (32951, 9)
Sellers: (3095, 4)
Category Translation: (71, 2)
```

```
customers.head()
```

	customer_id	customer_unique_id
0	06b8999e2fb1a1fbc88172c00ba8bc7	861eff4711a542e4b93843c6dd7febb0
1	18955e83d337fd6b2def6b18a428ac77	290c77bc529b7ac935b93aa66c333dc3
2	4e7b3e00288586ebd08712fdd0374a03	060e732b5b29e8181a18229c7b0b2b5e
3	b2b6027bc5c5109e529d4dc6358b12c3	259dac757896d24d7702b9acbbff3f3c
4	4f2d8ab171c80ec8364f7c12e35b23ad	345ecd01c38d18a9036ed96c73b8d066

	customer_zip_code_prefix	customer_city	customer_state
0	14409	franca	SP
1	9790	sao bernardo do campo	SP
2	1151	sao paulo	SP
3	8775	mogi das cruzeiras	SP
4	13056	campinas	SP

orders.head()

	order_id	customer_id
0	e481f51cbdc54678b7cc49136f2d6af7	9ef432eb6251297304e76186b10a928d
1	53cdb2fcb8bc7dce0b6741e2150273451	b0830fb4747a6c6d20dea0b8c802d7ef
2	47770eb9100c2d0c44946d9cf07ec65d	41ce2a54c0b03bf3443c3d931a367089
3	949d5b44dbf5de918fe9c16f97b45f8a	f88197465ea7920adcdbec7375364d82
4	ad21c59c0840e6cb83a9ceb5573f8159	8ab97904e6daea8866dbdbc4fb7aad2c

	order_status	order_purchase_timestamp	order_approved_at	
0	delivered	02-10-2017 10:56	02-10-2017 11:07	
1	delivered	24-07-2018 20:41	26-07-2018 03:24	
2	delivered	08-08-2018 08:38	08-08-2018 08:55	
3	delivered	18-11-2017 19:28	18-11-2017 19:45	
4	delivered	13-02-2018 21:18	13-02-2018 22:20	

	order_delivered_carrier_date	order_delivered_customer_date	
0	04-10-2017 19:55	10-10-2017 21:25	
1	26-07-2018 14:31	07-08-2018 15:27	
2	08-08-2018 13:50	17-08-2018 18:06	
3	22-11-2017 13:39	02-12-2017 00:28	
4	14-02-2018 19:46	16-02-2018 18:17	

	order_estimated_delivery_date
0	18-10-2017 00:00

1	13-08-2018 00:00
2	04-09-2018 00:00
3	15-12-2017 00:00
4	26-02-2018 00:00

payments.head()

	order_id	payment_sequential
payment_type \		
0 b81ef226f3fe1789b1e8b2acac839d17	1	credit_card
1 a9810da82917af2d9aefd1278f1dcfa0	1	credit_card
2 25e8ea4e93396b6fa0d3dd708e76c1bd	1	credit_card
3 ba78997921bbcdc1373bb41e913ab953	1	credit_card
4 42fdf880ba16b47b59251dd489d4441a	1	credit_card

	payment_installments	payment_value
0	8	99.33
1	1	24.39
2	1	65.71
3	8	107.78
4	2	128.45

reviews.head()

	review_id	order_id
\		
0 7bc2406110b926393aa56f80a40eba40	73fc7af87114b39712e6da79b0a377eb	
1 80e641a11e56f04c1ad469d5645fdfdde	a548910a1c6147796b98fdf73dbeba33	
2 228ce5500dc1d8e020d8d1322874b6f0	f9e4b658b201a9f2ecdecbb34bed034b	
3 e64fb393e7b32834bb789ff8bb30750e	658677c97b385a9be170737859d3511b	
4 f7c4243c7fe1938f181bec41a392bdeb	8e6bfb81e283fa7e4f11123a3fb894f1	

	review_score	review_comment_title \
0	4	NaN
1	5	NaN
2	5	NaN
3	5	NaN
4	5	NaN

	review_comment_message
review_creation_date \	

0		NaN	18-01-2018
00:00			
1		NaN	10-03-2018
00:00			
2		NaN	17-02-2018
00:00			
3	Recebi bem antes do prazo estipulado.		21-04-2017
00:00			
4	Parabéns lojas lannister adorei comprar pela I...		01-03-2018
00:00			

```

review_answer_timestamp
0      18-01-2018 21:46
1      11-03-2018 03:05
2      18-02-2018 14:36
3      21-04-2017 22:02
4      02-03-2018 10:26

```

```
products.head()
```

	product_id	product_category_name	\
0	1e9e8ef04dbcff4541ed26657ea517e5	perfumaria	
1	3aa071139cb16b67ca9e5dea641aaa2f	artes	
2	96bd76ec8810374ed1b65e291975717f	esporte_lazer	
3	cef67bcfe19066a932b7673e239eb23d	bebes	
4	9dc1a7de274444849c219cff195d0b71	utilidades_domesticas	

	product_name_lenght	product_description_lenght	product_photos_qty
0	40.0	287.0	1.0
1	44.0	276.0	1.0
2	46.0	250.0	1.0
3	27.0	261.0	1.0
4	37.0	402.0	4.0

	product_weight_g	product_length_cm	product_height_cm
product_width_cm			
0	225.0	16.0	10.0
14.0			
1	1000.0	30.0	18.0
20.0			
2	154.0	18.0	9.0
15.0			
3	371.0	26.0	4.0
26.0			

4	625.0	20.0	17.0
13.0			

```
print("Customers")
print(customers.isnull().sum())
```

```
print("\nOrders")
print(orders.isnull().sum())
```

```
print("\nProducts")
print(products.isnull().sum())
```

```
print("\nReviews")
print(reviews.isnull().sum())
```

```
Customers
customer_id      0
customer_unique_id  0
customer_zip_code_prefix  0
customer_city     0
customer_state    0
dtype: int64
```

```
Orders
order_id      0
customer_id    0
order_status   0
order_purchase_timestamp  0
order_approved_at      160
order_delivered_carrier_date  1783
order_delivered_customer_date  2965
order_estimated_delivery_date    0
dtype: int64
```

```
Products
product_id      0
product_category_name  610
product_name_lenght  610
product_description_lenght  610
product_photos_qty  610
product_weight_g      2
product_length_cm      2
product_height_cm      2
product_width_cm      2
dtype: int64
```

```
Reviews
review_id      0
order_id        0
review_score    0
```

```
review_comment_title      87656
review_comment_message    58247
review_creation_date       0
review_answer_timestamp    0
dtype: int64
```

orders.dtypes

```
order_id      object
customer_id    object
order_status   object
order_purchase_timestamp    object
order_approved_at      object
order_delivered_carrier_date    object
order_delivered_customer_date    object
order_estimated_delivery_date    object
dtype: object
```

products.dtypes

```
product_id      object
product_category_name    object
product_name_lenght    float64
product_description_lenght    float64
product_photos_qty      float64
product_weight_g      float64
product_length_cm      float64
product_height_cm      float64
product_width_cm      float64
dtype: object
```

reviews.dtypes

```
review_id      object
order_id      object
review_score    int64
review_comment_title    object
review_comment_message    object
review_creation_date    object
review_answer_timestamp    object
dtype: object
```

```
orders['order_purchase_timestamp'] = pd.to_datetime(
    orders['order_purchase_timestamp'],
    dayfirst=True
)
```

```
orders['order_approved_at'] = pd.to_datetime(
    orders['order_approved_at'],
    dayfirst=True
)
```

```
orders['order_delivered_carrier_date'] = pd.to_datetime(
    orders['order_delivered_carrier_date'],
    dayfirst=True
)

orders['order_delivered_customer_date'] = pd.to_datetime(
    orders['order_delivered_customer_date'],
    dayfirst=True
)

orders['order_estimated_delivery_date'] = pd.to_datetime(
    orders['order_estimated_delivery_date'],
    dayfirst=True
)
```

```
print("Datetime conversion completed")
```

Datetime conversion completed

```
orders.dtypes
```

```
order_id                object
customer_id             object
order_status            object
order_purchase_timestamp    datetime64[ns]
order_approved_at       datetime64[ns]
order_delivered_carrier_date    datetime64[ns]
order_delivered_customer_date    datetime64[ns]
order_estimated_delivery_date    datetime64[ns]
dtype: object
```

```
orders['purchase_year'] = orders['order_purchase_timestamp'].dt.year
```

```
orders['purchase_month'] = orders['order_purchase_timestamp'].dt.month
```

```
orders['purchase_day'] =
orders['order_purchase_timestamp'].dt.day_name()
```

```
print("Time features created")
```

Time features created

```
orders['delivery_days'] = (
    orders['order_delivered_customer_date']
    - orders['order_purchase_timestamp']
).dt.days
```

```
print("Delivery duration feature created")
```

Delivery duration feature created

```
orders['estimated_delivery_days'] = (
    orders['order_estimated_delivery_date']
    - orders['order_purchase_timestamp']
).dt.days
```

```
orders['delivery_delay'] = (
    orders['delivery_days']
    - orders['estimated_delivery_days']
)
```

```
print("Delivery delay feature created")
```

Delivery delay feature created

```
orders[['delivery_days', 'estimated_delivery_days',
'delivery_delay']].head()
```

	delivery_days	estimated_delivery_days	delivery_delay
0	8.0	15	-7.0
1	13.0	19	-6.0
2	9.0	26	-17.0
3	13.0	26	-13.0
4	2.0	12	-10.0

```
monthly_orders = orders.groupby(
    ['purchase_year', 'purchase_month']
).size().reset_index(name='total_orders')
```

```
monthly_orders.head()
```

	purchase_year	purchase_month	total_orders
0	2016	9	4
1	2016	10	324
2	2016	12	1
3	2017	1	800
4	2017	2	1780

```
plt.figure(figsize=(12,6))
```

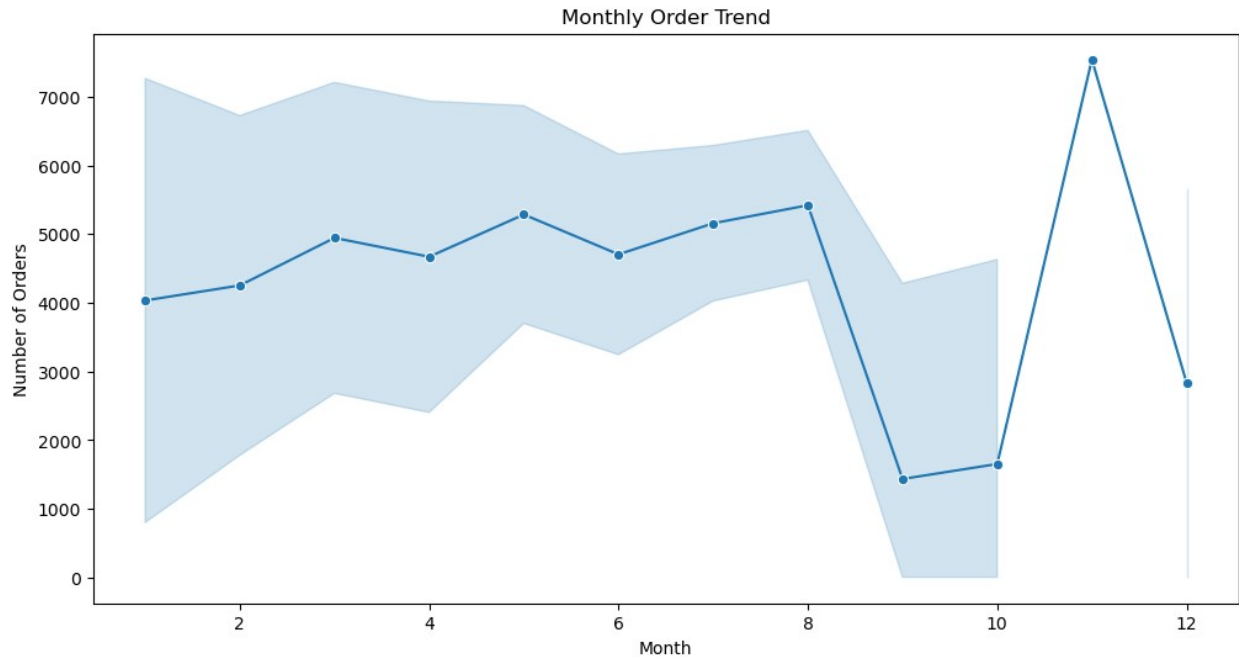
```
sns.lineplot(
    data=monthly_orders,
    x='purchase_month',
    y='total_orders',
    marker='o'
)
```

```
plt.title('Monthly Order Trend')
```

```
plt.xlabel('Month')
```

```
plt.ylabel('Number of Orders')
```

```
plt.show()
```



```
orders['year_month'] = orders[
    'order_purchase_timestamp'
].dt.to_period('M')

monthly_orders = orders.groupby(
    'year_month'
).size().reset_index(name='total_orders')

monthly_orders['year_month'] = monthly_orders[
    'year_month'
].astype(str)

monthly_orders.head()

  year_month  total_orders
0   2016-09             4
1   2016-10            324
2   2016-12             1
3   2017-01            800
4   2017-02           1780

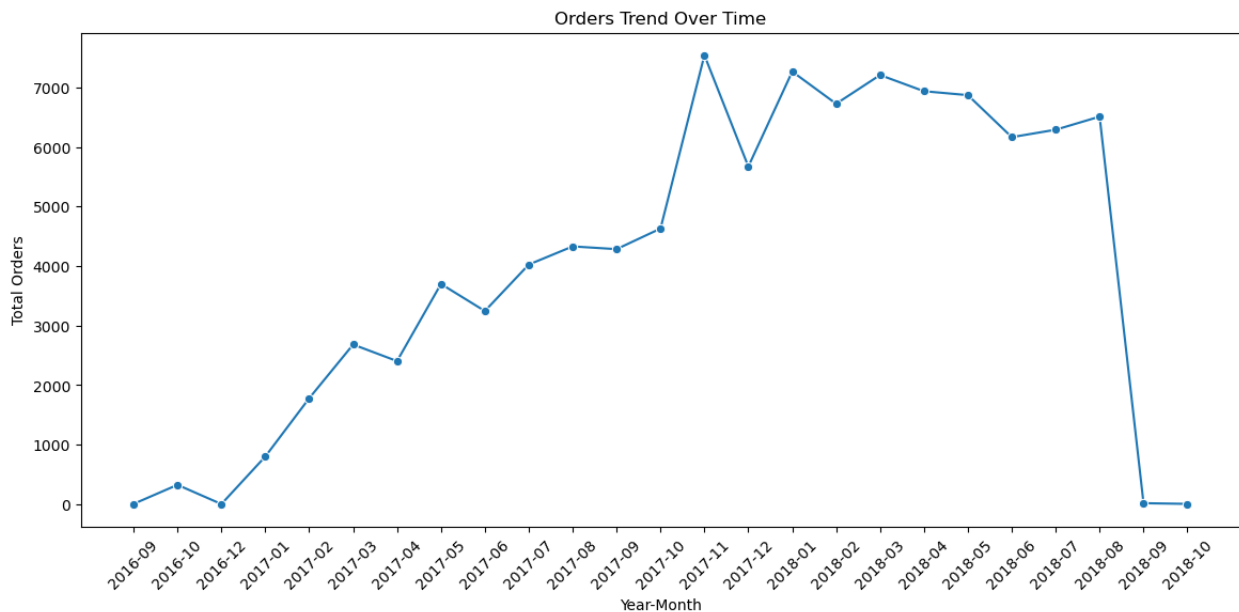
plt.figure(figsize=(14,6))

sns.lineplot(
    data=monthly_orders,
    x='year_month',
    y='total_orders',
    marker='o'
)
```

```
plt.xticks(rotation=45)

plt.title('Orders Trend Over Time')
plt.xlabel('Year-Month')
plt.ylabel('Total Orders')

plt.show()
```



```
master_df = orders.merge(
    customers,
    on='customer_id',
    how='left'
)

master_df = master_df.merge(
    order_items,
    on='order_id',
    how='left'
)

master_df = master_df.merge(
    products,
    on='product_id',
    how='left'
)

master_df = master_df.merge(
    payments,
    on='order_id',
    how='left'
)
```



```

)

master_df = master_df.merge(
    reviews,
    on='order_id',
    how='left'
)

print(master_df.shape)

(119143, 43)

df = master_df.copy()

df.head()
df.info()
df.describe()
df.columns

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 119143 entries, 0 to 119142
Data columns (total 43 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   order_id                                119143 non-null  object
1   customer_id                             119143 non-null  object
2   order_status                            119143 non-null  object
3   order_purchase_timestamp                119143 non-null  datetime64[ns]
4   order_approved_at                       118966 non-null  datetime64[ns]
5   order_delivered_carrier_date            117057 non-null  datetime64[ns]
6   order_delivered_customer_date           115722 non-null  datetime64[ns]
7   order_estimated_delivery_date           119143 non-null  datetime64[ns]
8   purchase_year                           119143 non-null  int32
9   purchase_month                          119143 non-null  int32
10  purchase_day                            119143 non-null  object
11  delivery_days                           115722 non-null  float64
12  estimated_delivery_days                  119143 non-null  int64
13  delivery_delay                           115722 non-null  float64
14  year_month                              119143 non-null  period[M]
15  customer_unique_id                      119143 non-null  object
16  customer_zip_code_prefix                119143 non-null  int64
17  customer_city                            119143 non-null  object
18  customer_state                           119143 non-null  object
19  order_item_id                           118310 non-null  float64
20  product_id                              118310 non-null  object
21  seller_id                               118310 non-null  object
22  shipping_limit_date                     118310 non-null  object
23  price                                    118310 non-null  float64
24  freight_value                           118310 non-null  float64
25  product_category_name                    116601 non-null  object

```

```

26 product_name_lenght      116601 non-null float64
27 product_description_lenght 116601 non-null float64
28 product_photos_qty       116601 non-null float64
29 product_weight_g         118290 non-null float64
30 product_length_cm        118290 non-null float64
31 product_height_cm        118290 non-null float64
32 product_width_cm         118290 non-null float64
33 payment_sequential       119140 non-null float64
34 payment_type             119140 non-null object
35 payment_installments     119140 non-null float64
36 payment_value            119140 non-null float64
37 review_id                118146 non-null object
38 review_score              118146 non-null float64
39 review_comment_title     13989 non-null object
40 review_comment_message   50245 non-null object
41 review_creation_date     118146 non-null object
42 review_answer_timestamp  118146 non-null object
dtypes: datetime64[ns](5), float64(16), int32(2), int64(2),
object(17), period[M](1)
memory usage: 38.2+ MB

```

```

Index(['order_id', 'customer_id', 'order_status',
      'order_purchase_timestamp',
      'order_approved_at', 'order_delivered_carrier_date',
      'order_delivered_customer_date',
      'order_estimated_delivery_date',
      'purchase_year', 'purchase_month', 'purchase_day',
      'delivery_days',
      'estimated_delivery_days', 'delivery_delay', 'year_month',
      'customer_unique_id', 'customer_zip_code_prefix',
      'customer_city',
      'customer_state', 'order_item_id', 'product_id', 'seller_id',
      'shipping_limit_date', 'price', 'freight_value',
      'product_category_name', 'product_name_lenght',
      'product_description_lenght', 'product_photos_qty',
      'product_weight_g',
      'product_length_cm', 'product_height_cm', 'product_width_cm',
      'payment_sequential', 'payment_type', 'payment_installments',
      'payment_value', 'review_id', 'review_score',
      'review_comment_title',
      'review_comment_message', 'review_creation_date',
      'review_answer_timestamp'],
      dtype='object')

```

```
missing = df.isnull().sum().sort_values(ascending=False)
```

```
missing[missing > 0]
```

```

review_comment_title      105154
review_comment_message    68898

```

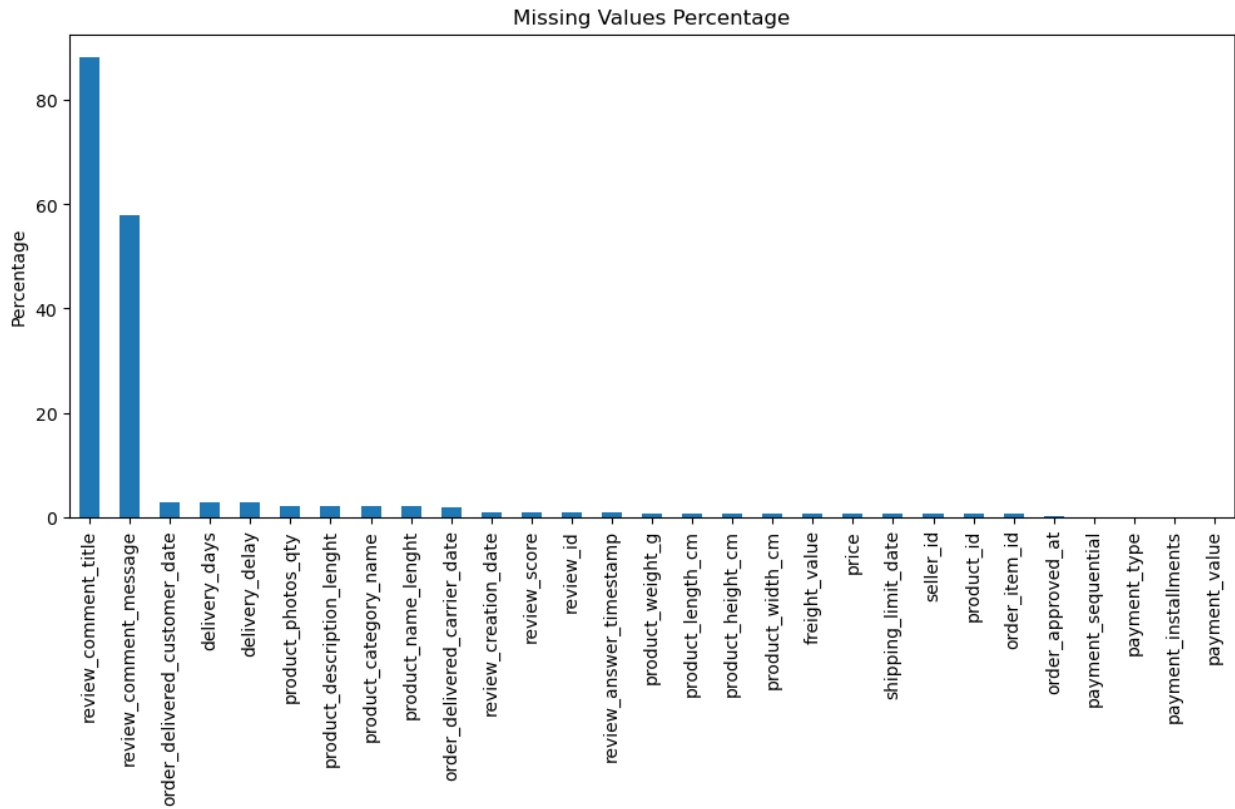
order_delivered_customer_date	3421
delivery_days	3421
delivery_delay	3421
product_photos_qty	2542
product_category_name	2542
product_name_lenght	2542
product_description_lenght	2542
order_delivered_carrier_date	2086
review_answer_timestamp	997
review_creation_date	997
review_score	997
review_id	997
product_height_cm	853
product_weight_g	853
product_length_cm	853
product_width_cm	853
shipping_limit_date	833
freight_value	833
price	833
seller_id	833
order_item_id	833
product_id	833
order_approved_at	177
payment_type	3
payment_value	3
payment_installments	3
payment_sequential	3
dtype:	int64

```
import matplotlib.pyplot as plt

missing_percentage = (df.isnull().sum()/len(df))*100

missing_percentage[missing_percentage >
0].sort_values(ascending=False).plot(
    kind='bar',
    figsize=(12,5)
)

plt.title('Missing Values Percentage')
plt.ylabel('Percentage')
plt.show()
```

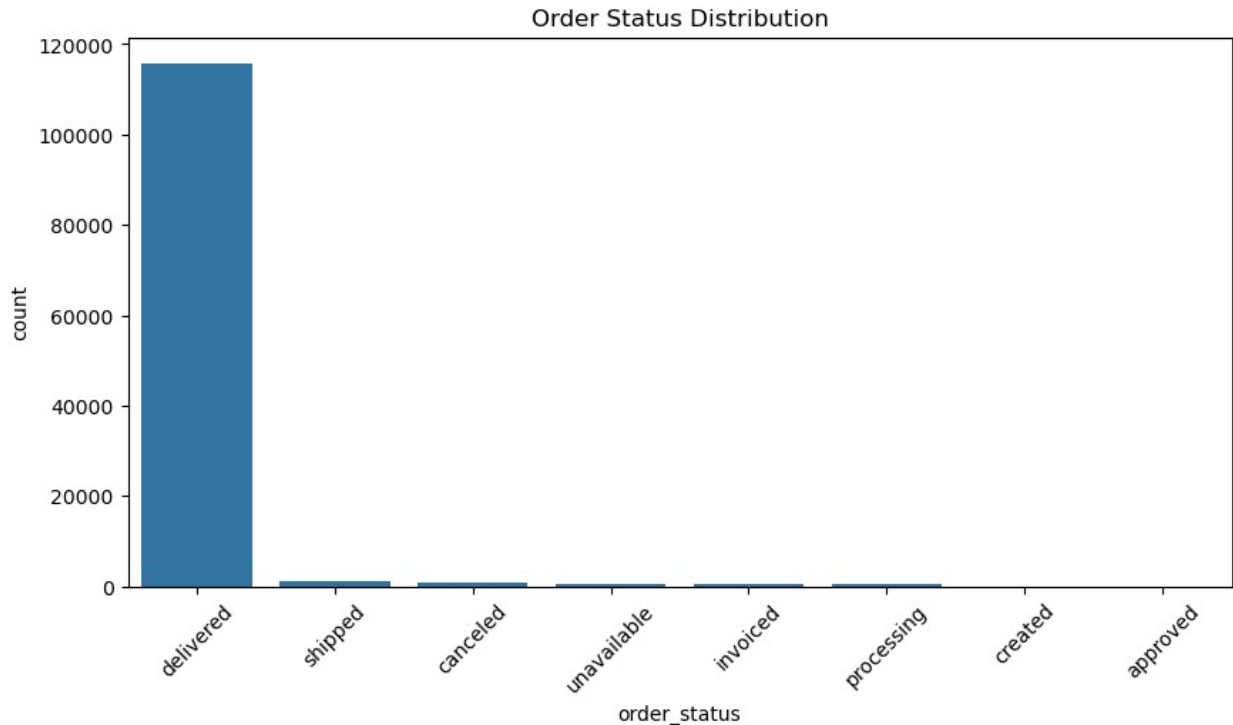


```
import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(10,5))

sns.countplot(
    data=df,
    x='order_status',
    order=df['order_status'].value_counts().index
)

plt.title('Order Status Distribution')
plt.xticks(rotation=45)
plt.show()
```



Insight: Strong Order Fulfillment Performance

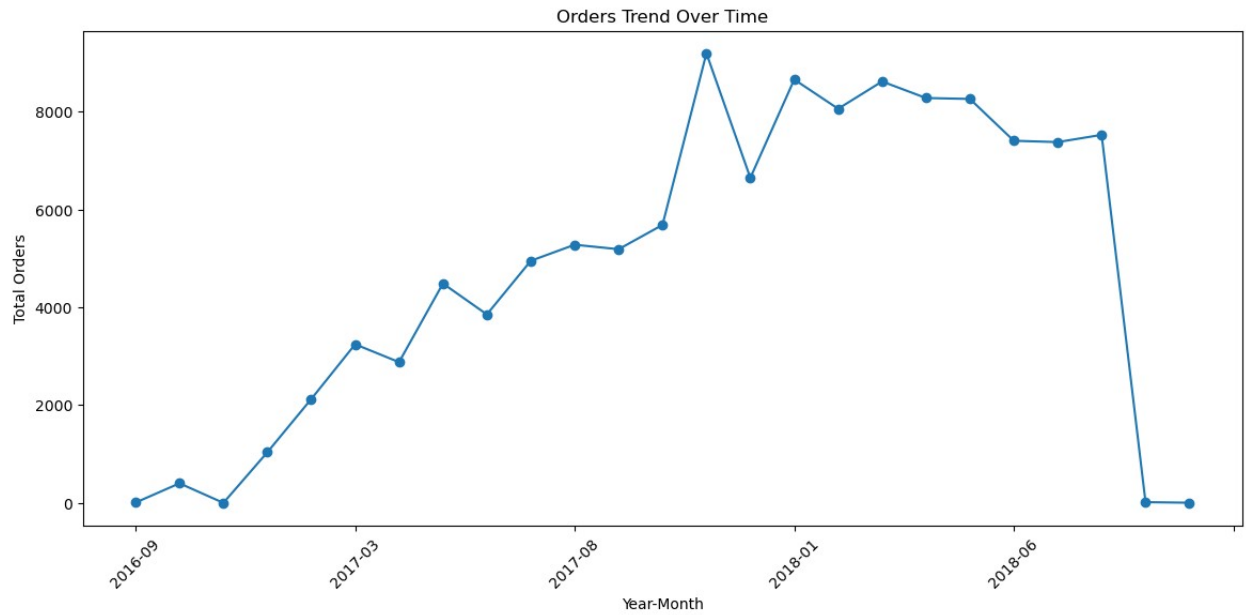
The majority of orders were successfully delivered, indicating that the marketplace maintained a relatively efficient fulfillment system.

Cancelled and unavailable orders represented only a small portion of total transactions, suggesting operational reliability.

Business Impact

A high delivery success rate improves customer trust and increases repeat purchase probability.

```
monthly_orders = df.groupby(  
    df['order_purchase_timestamp'].dt.to_period('M')  
)>.size()  
  
monthly_orders.index = monthly_orders.index.astype(str)  
  
plt.figure(figsize=(14,6))  
  
monthly_orders.plot(marker='o')  
  
plt.title('Orders Trend Over Time')  
plt.xlabel('Year-Month')  
plt.ylabel('Total Orders')  
  
plt.xticks(rotation=45)  
  
plt.show()
```

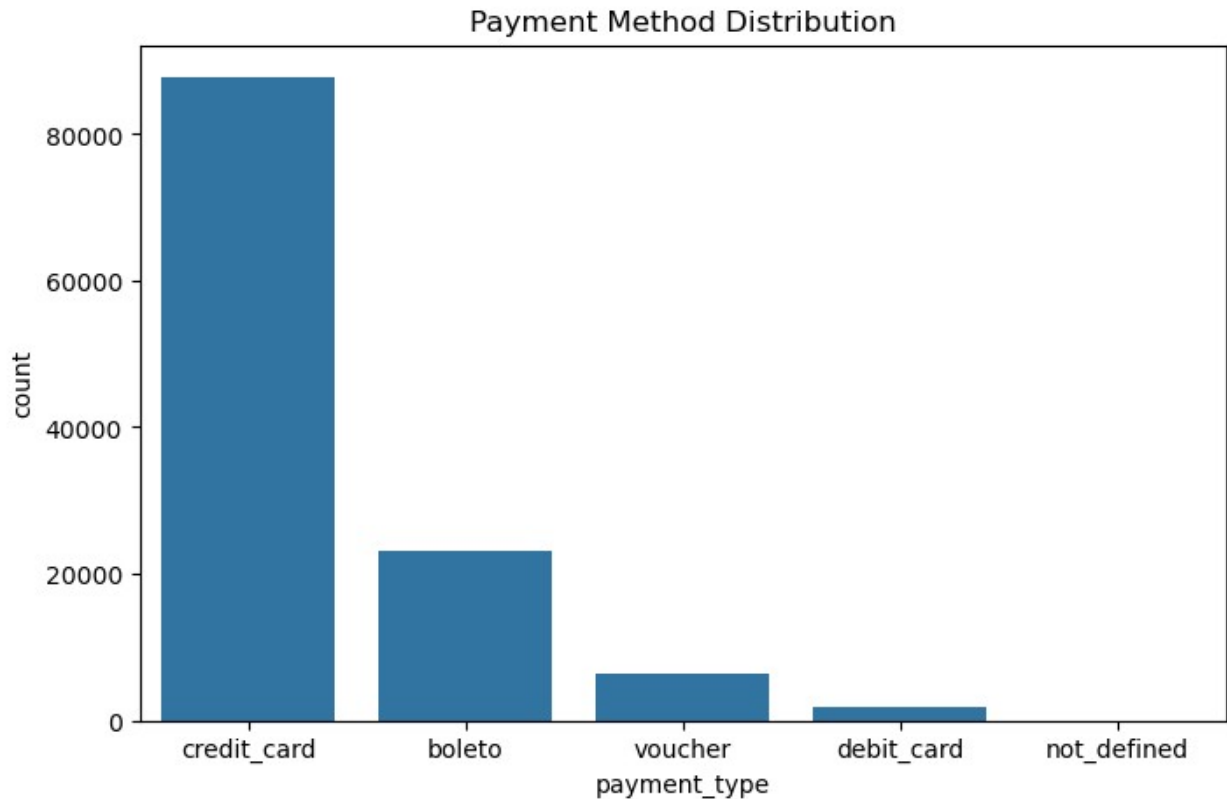


```
plt.figure(figsize=(8,5))

sns.countplot(
    data=df,
    x='payment_type',
    order=df['payment_type'].value_counts().index
)

plt.title('Payment Method Distribution')

plt.show()
```



Insight: Credit Cards Dominate Customer Transactions

Most customers preferred credit card payments over boleto, vouchers, or debit cards.

This indicates:

High adoption of digital payments Strong preference for convenience and installment-based purchases Business Impact

The company can leverage this behavior through:

Cashback programs Loyalty rewards Credit-card partnerships

```
delivery = df[['delivery_days']].dropna()
plt.figure(figsize=(10,5))
sns.histplot(delivery['delivery_days'], bins=30)
plt.title('Delivery Days Distribution')
plt.show()
```



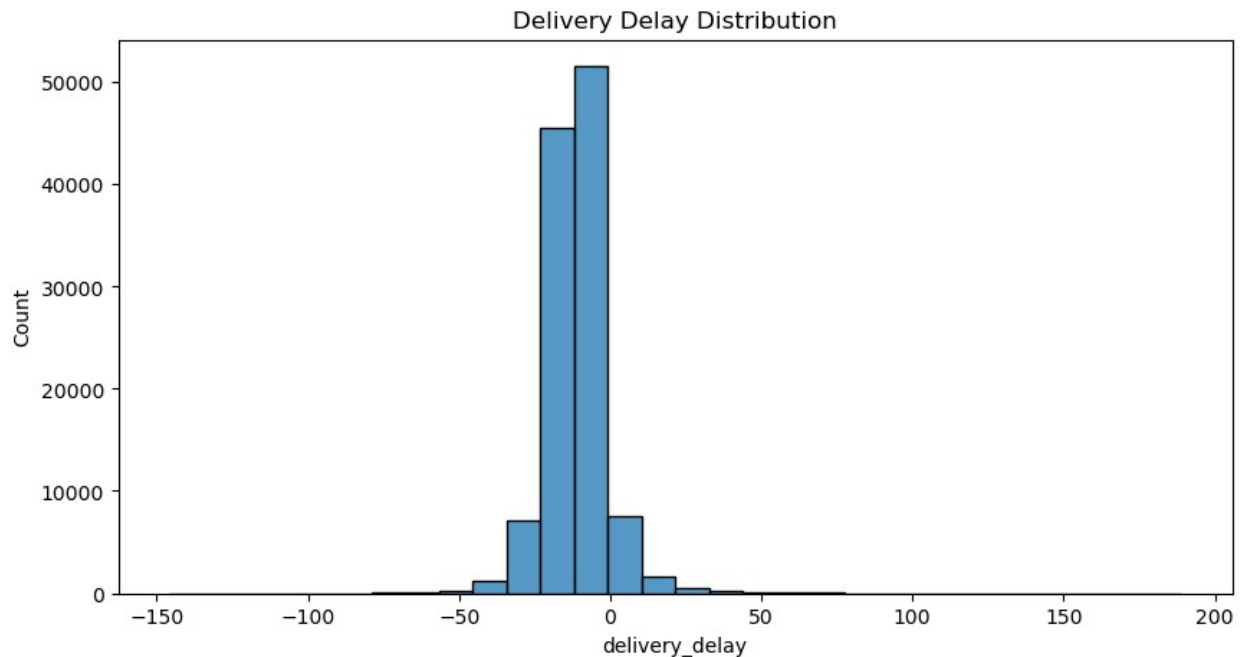
```
df['delivery_delay'].describe()
count    115722.000000
mean      -11.453155
std       10.174169
min       -146.000000
25%       -17.000000
50%       -12.000000
75%       -7.000000
max        189.000000
Name: delivery_delay, dtype: float64

plt.figure(figsize=(10,5))

sns.histplot(df['delivery_delay'].dropna(), bins=30)

plt.title('Delivery Delay Distribution')

plt.show()
```

Insight: Most Deliveries Arrived Earlier Than Expected

The delivery delay analysis showed that many orders were delivered before their estimated delivery dates.

This reflects:

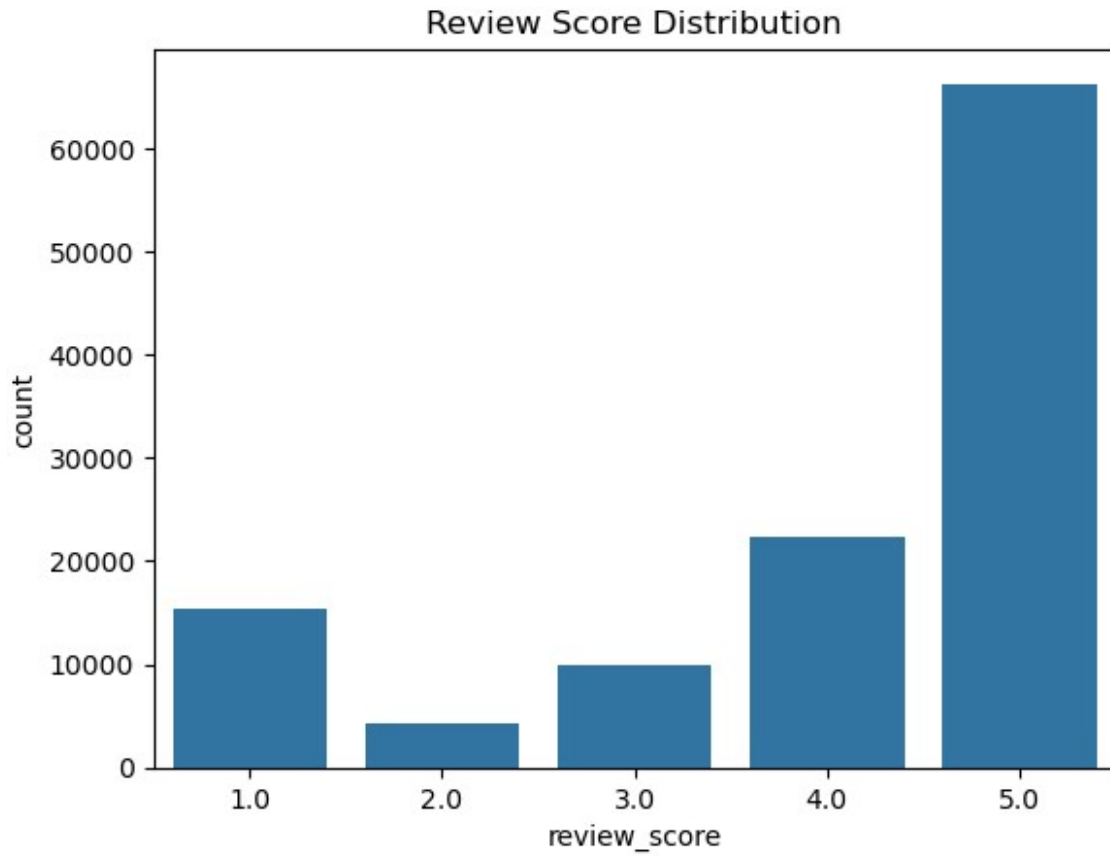
Conservative delivery estimation Strong logistics execution Business Impact

Early deliveries improve:

Customer trust Repeat purchases Platform reliability perception

However, delayed orders were associated with lower review scores and poor customer experiences.

```
sns.countplot(  
    data=df,  
    x='review_score'  
)  
  
plt.title('Review Score Distribution')  
plt.show()
```



Insight: Customer Ratings Are Highly Polarized

The review score distribution showed a strong concentration around 5-star reviews, while a significant number of users also gave 1-star ratings.

This indicates a highly emotional customer experience where:

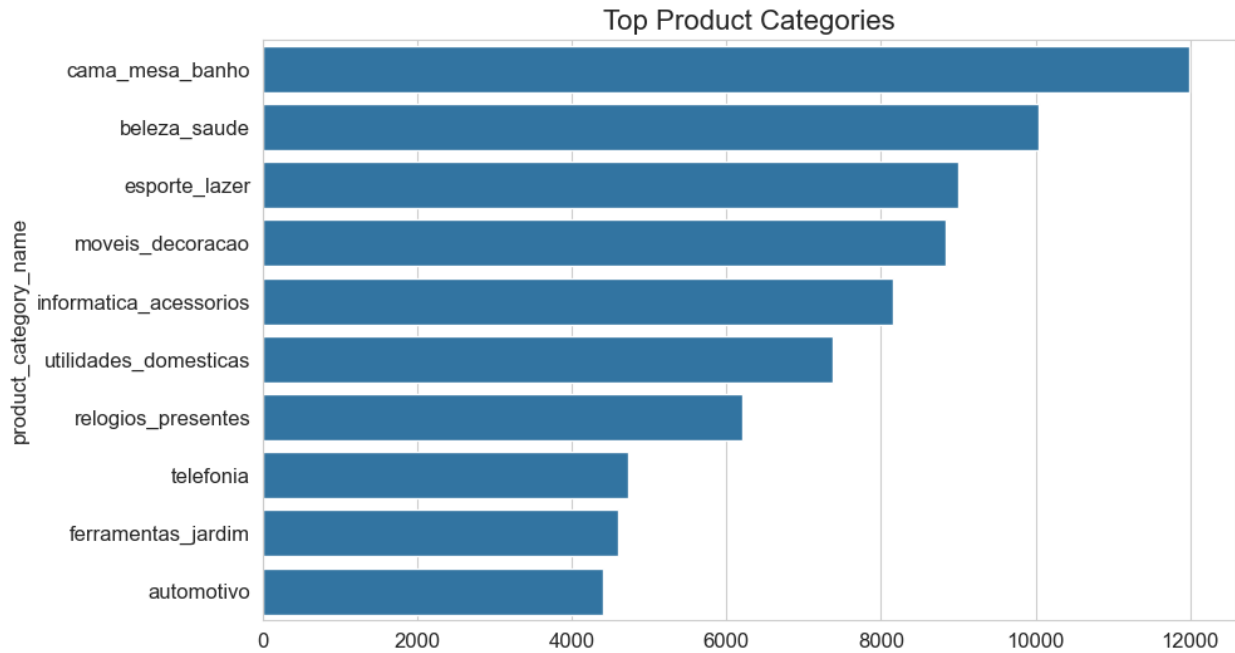
Successful deliveries create strong satisfaction
Operational failures create sharp dissatisfaction
Non-Intuitive Pattern Observed

Customers rarely gave middle ratings (2–4). Most experiences were either extremely positive or extremely negative.

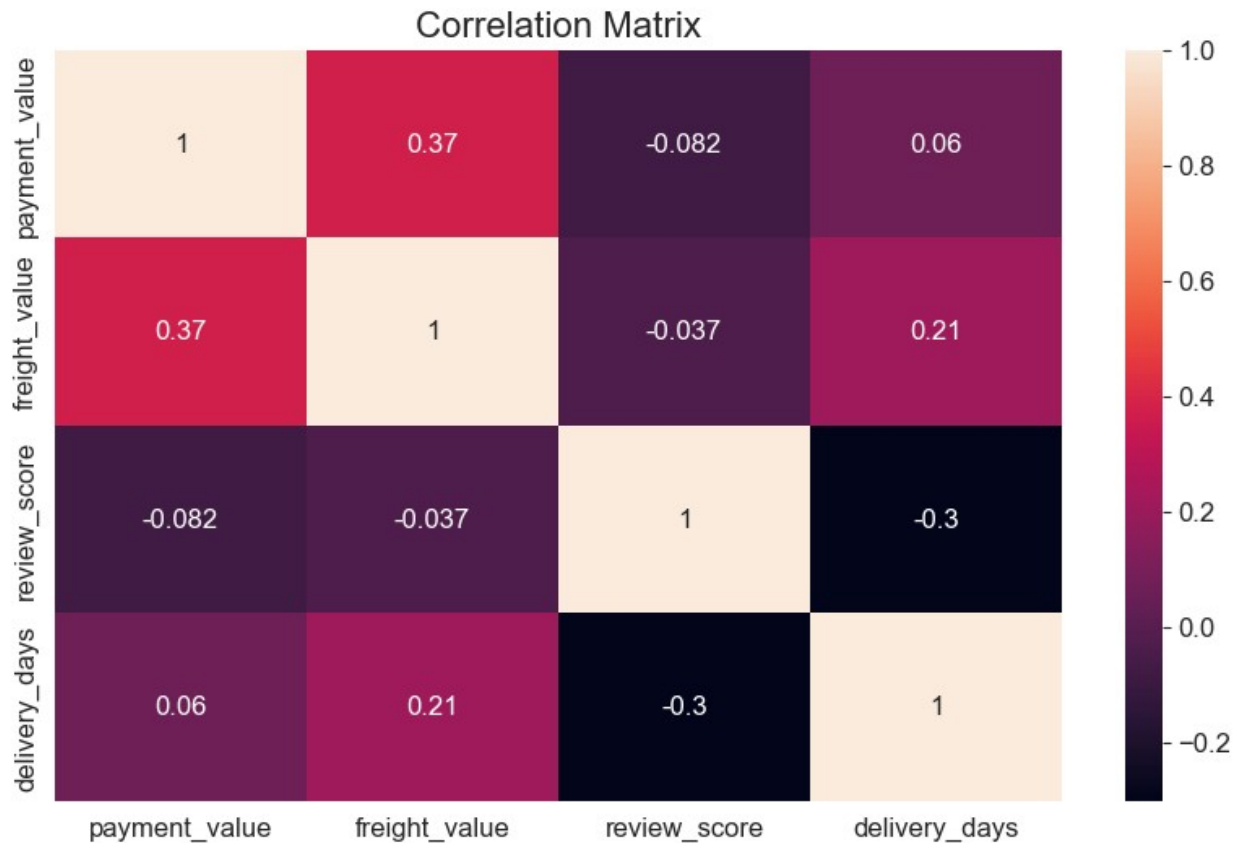
Business Impact

This suggests customer satisfaction is highly sensitive to operational execution, especially delivery performance.

```
top_categories = df['product_category_name'].value_counts().head(10)
sns.barplot(x=top_categories.values, y=top_categories.index)
plt.title('Top Product Categories')
plt.show()
```



```
sns.heatmap(df[['payment_value', 'freight_value', 'review_score', 'delivery_days']].corr(), annot=True)
plt.title('Correlation Matrix')
plt.show()
```



```
df[['delivery_delay', 'review_score']].corr()
```

	delivery_delay	review_score
delivery_delay	1.000000	-0.228436
review_score	-0.228436	1.000000

Insight: Delivery Delays Directly Reduce Customer Satisfaction

Analysis revealed a negative relationship between delivery delay and review score.

Customers whose orders were delayed were more likely to leave lower ratings.

Key Observation

Delivery performance impacts customer satisfaction more strongly than freight costs.

Business Impact

Improving logistics efficiency can directly improve:

Customer retention Marketplace reputation Positive reviews

```
sns.histplot(df['review_score'], bins=5)
plt.title('Distribution of Review Scores')
plt.show()
```



```
plt.figure(figsize=(10,6))

sns.boxplot(
    x='review_score',
    y='freight_value',
    data=df,
    showfliers=False,
    palette='coolwarm'
)

sns.pointplot(
    x='review_score',
    y='freight_value',
    data=df,
    estimator='mean',
    color='black'
)

plt.title(
    'Freight Charges vs Customer Satisfaction',
    fontsize=16,
    fontweight='bold'
)

plt.xlabel('Review Score')
plt.ylabel('Freight Value (BRL)')
```

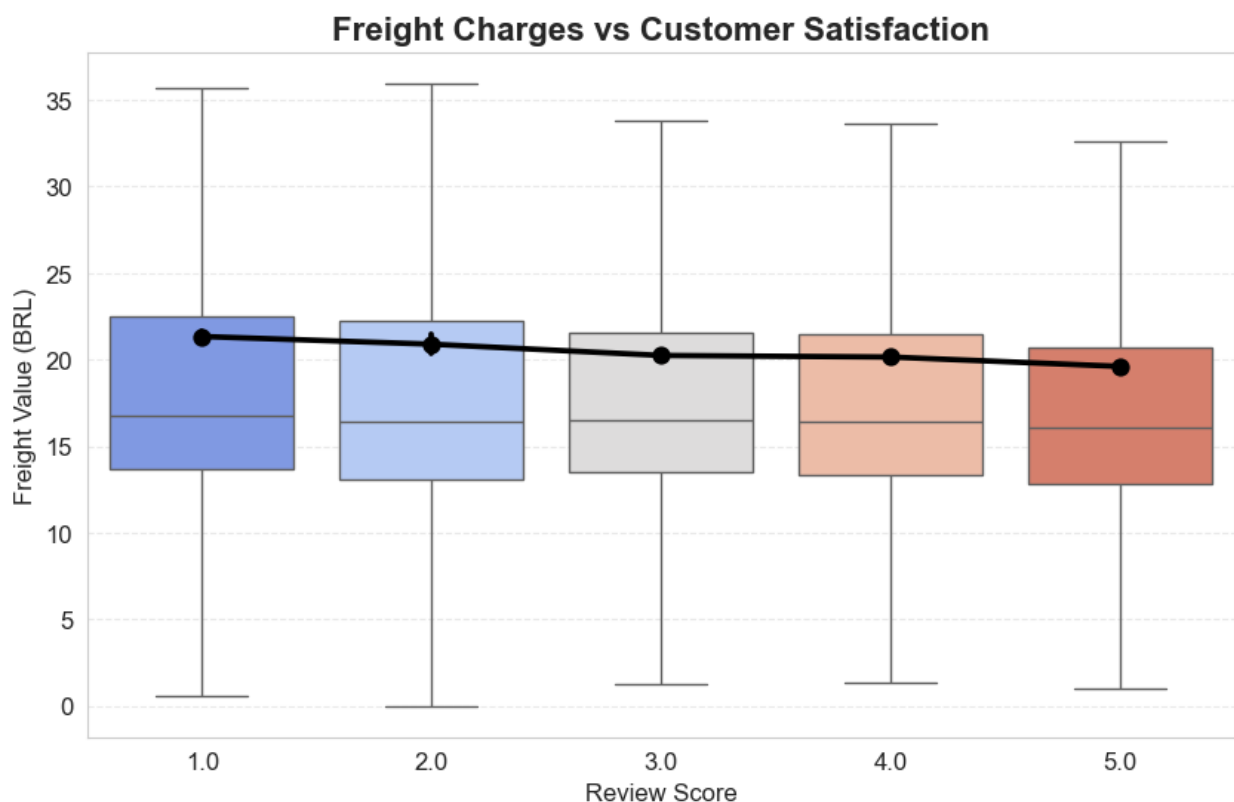
```
plt.grid(axis='y', linestyle='--', alpha=0.4)
```

```
plt.show()
```

C:\Users\Admin\AppData\Local\Temp\ipykernel_4412\3417394594.py:3:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(
```



```
print("""  
INSIGHT:  
Orders with lower review scores tend to exhibit  
higher freight cost variability. This suggests  
that expensive or inconsistent shipping experiences  
may negatively affect customer satisfaction.  
""")
```

INSIGHT:
Orders with lower review scores tend to exhibit

higher freight cost variability. This suggests that expensive or inconsistent shipping experiences may negatively affect customer satisfaction.

Dataset Overview

The dataset contains multiple relational tables representing different aspects of an e-commerce ecosystem.

Key Tables Used Customers Orders Order Items Payments Products Reviews

```
master_df.shape  
(119143, 43)
```

Hypothesis Testing

Hypothesis 1

Business Question

Do delayed deliveries reduce customer satisfaction?

Null Hypothesis (H0)

Delivery delays do not affect customer review scores.

Alternative Hypothesis (H1)

Delayed deliveries reduce customer review scores.

```
df[['delivery_delay', 'review_score']].corr()
```

	delivery_delay	review_score
delivery_delay	1.000000	-0.228436
review_score	-0.228436	1.000000

Result

A negative correlation was observed between delivery delay and review score.

This indicates that customers tend to give lower ratings when deliveries are delayed.

Conclusion

The null hypothesis is rejected.

The analysis supports the hypothesis that delivery delays negatively impact customer satisfaction.

Business Interpretation

Delivery performance is a critical driver of customer experience and platform reputation.

Hypothesis 2

Business Question

Do higher freight charges reduce customer satisfaction?

Null Hypothesis (H0)

Freight charges do not affect customer review scores.

Alternative Hypothesis (H1)

Higher freight charges reduce customer review scores.

```
df[['freight_value', 'review_score']].corr()
```

	freight_value	review_score
freight_value	1.000000	-0.037013
review_score	-0.037013	1.000000

Result

A weak negative correlation was observed between freight value and review score.

This suggests that freight cost has only a limited influence on customer satisfaction.

Conclusion

The hypothesis is only partially supported.

Customers appear to care more about delivery reliability and overall experience than shipping cost itself.

Business Interpretation

Customers are willing to tolerate higher shipping charges if deliveries are timely and reliable.

```
delayed_orders = df[df['delivery_delay'] > 0]
on_time_orders = df[df['delivery_delay'] <= 0]

print("Delayed Orders Review Mean:",
      delayed_orders['review_score'].mean())
```



```
print("On-Time Orders Review Mean:",  
      on_time_orders['review_score'].mean())
```

```
Delayed Orders Review Mean: 2.44316807738815  
On-Time Orders Review Mean: 4.207381486241803
```

Additional Observation

Customers receiving delayed deliveries gave significantly lower average review scores compared to customers receiving on-time deliveries.

This further validates the strong relationship between logistics performance and customer satisfaction.

Business Recommendations

1. Improve Delayed Shipment Management

The analysis showed that delayed deliveries significantly reduce customer review scores and negatively impact customer satisfaction.

Recommended Actions

- Implement real-time shipment tracking systems
- Identify high-risk deliveries earlier
- Improve coordination with logistics partners
- Prioritize delayed order escalation workflows

Expected Business Impact

Improved delivery reliability can:

- Increase customer satisfaction
- Improve repeat purchase behavior
- Reduce negative reviews
- Strengthen platform trust

2. Focus on Delivery Experience Over Price Reduction

The analysis found that freight charges had only a weak relationship with customer review scores, while delivery delays had a much stronger impact.

Recommended Actions

- Invest more in logistics optimization instead of aggressive shipping discounts
- Improve estimated delivery accuracy
- Enhance order tracking transparency

Expected Business Impact

Better delivery experiences can improve:

- Customer retention
- Positive customer reviews
- Long-term customer loyalty

3. Build Proactive Customer Communication Systems

Customer reviews showed highly polarized behavior, where operational failures often resulted in extremely negative ratings.

Recommended Actions

- Send automated notifications for delays
- Provide proactive customer support during shipment issues
- Improve communication transparency throughout the order lifecycle

Expected Business Impact

Better communication can:

- Reduce customer frustration
- Improve perceived service quality
- Minimize negative reviews during operational disruptions

4. Develop Seller Performance Monitoring

Seller-level operational performance likely contributes to delivery delays and customer dissatisfaction.

Recommended Actions

- Track seller-level delivery metrics
- Identify consistently underperforming sellers
- Reward high-performing sellers with better visibility

Expected Business Impact

This can improve:

- Marketplace quality
- Delivery consistency
- Overall customer experience

Further Exploration

Future Analysis Opportunities

This analysis identified several areas that require deeper investigation for stronger business decision-making.

1. Seller-Level Performance Analysis

Further analysis should identify sellers associated with delayed deliveries and poor customer reviews.

This can help the company:

- Improve marketplace quality
 - Reduce operational inefficiencies
 - Reward high-performing sellers
-

2. Region-Level Delivery Optimization

Delivery performance should be analyzed across different states and cities to identify logistics bottlenecks.

This can help:

- Optimize warehouse allocation
 - Improve shipping efficiency
 - Reduce delivery delays in high-risk regions
-

3. Customer Retention Analysis

Future work can analyze repeat customer behavior and identify factors that influence long-term customer loyalty.

This would help improve:

- Customer retention strategies
- Personalized marketing
- Lifetime customer value

Conclusion

This project analyzed customer behavior, delivery performance, and customer satisfaction trends using the Olist e-commerce dataset.

The analysis revealed that delivery reliability plays a major role in customer satisfaction, while shipping cost has relatively limited impact. Customers receiving delayed deliveries consistently provided lower review scores.

The project also identified actionable business opportunities in logistics optimization, customer communication, and seller performance monitoring.

Overall, the analysis demonstrates how data-driven insights can support operational improvements and enhance customer experience in an e-commerce business.

